

TP 9 : Apache kafka

Université / École :

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Installation du Kafka

1. Téléchargement et extraction des fichiers

```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...
nvim env.zsh

) wget https://downloads.apache.org/kafka/4.2.0/kafka_2.13-4.2.0.tgz
--2026-04-29 10:31:59-- https://downloads.apache.org/kafka/4.2.0/kafka_2.13-4.2.0.tgz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: 136097050 (130M) [application/x-gzip]
Saving to: 'kafka_2.13-4.2.0.tgz'

kafka_2.13-4.2.0.tg 100%[=====>] 129.79M  1.37MB/s   in 3m 32s

2026-04-29 10:35:32 (627 KB/s) - 'kafka_2.13-4.2.0.tgz' saved [136097050/136097050]

) mkdir ~/kafka
) tar -xvf kafka_2.13-4.2.0.tgz -C ~/kafka
kafka_2.13-4.2.0/
kafka_2.13-4.2.0/LICENSE
kafka_2.13-4.2.0/NOTICE
kafka_2.13-4.2.0/bin/
kafka_2.13-4.2.0/bin/kafka-delete-records.sh
kafka_2.13-4.2.0/bin/trogdor.sh
```

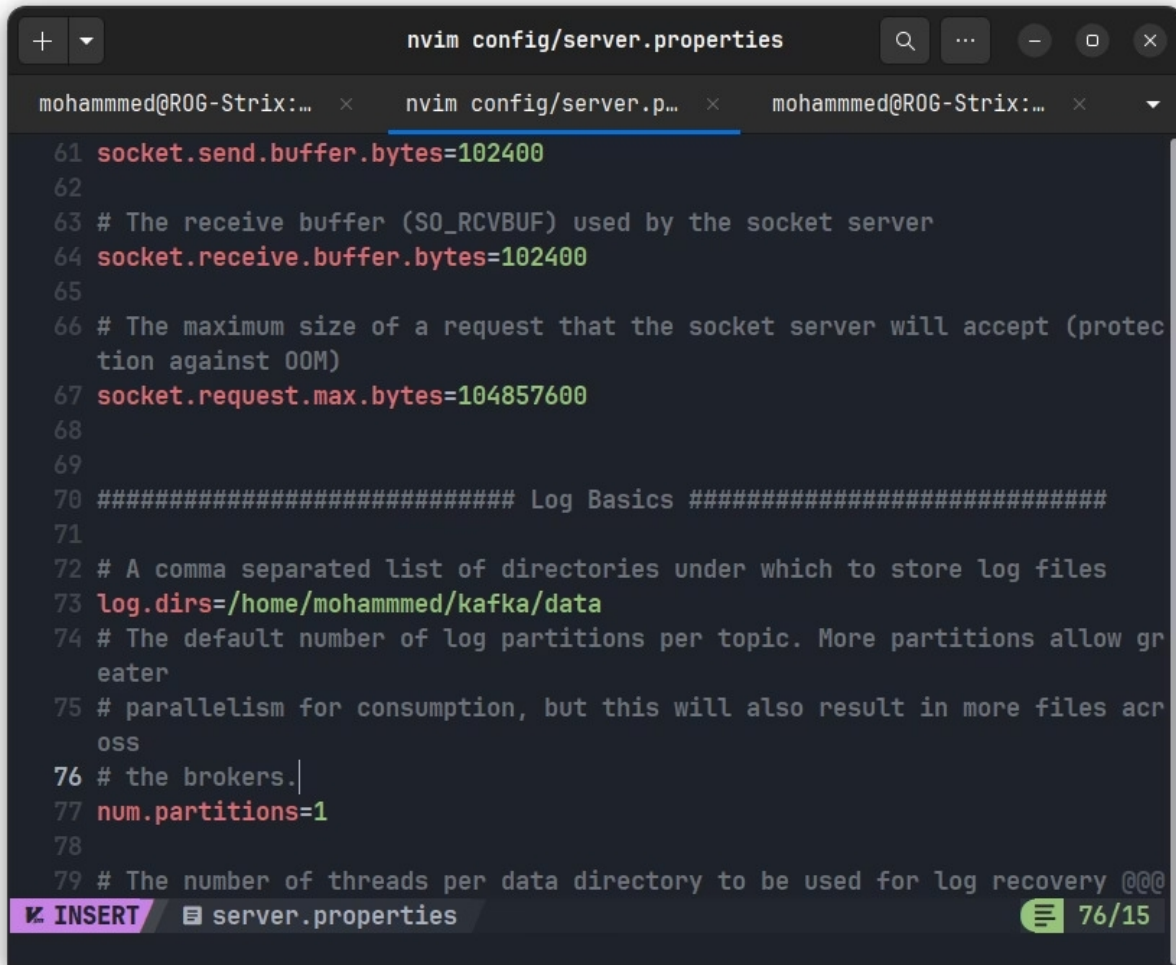
2. Configuration des variable d'environnement

```
nvim env.zsh

mohammed@ROG-Strix:~/Desktop/COUR...
nvim env.zsh

37 export HBASE_HOME=$HOME/hbase
38 export PATH=$PATH:$HBASE_HOME/bin
39
40 # Kafka configuration
41 export KAFKA_HOME=$HOME/kafka
42 export PATH=$PATH:$KAFKA_HOME/bin |
43
```

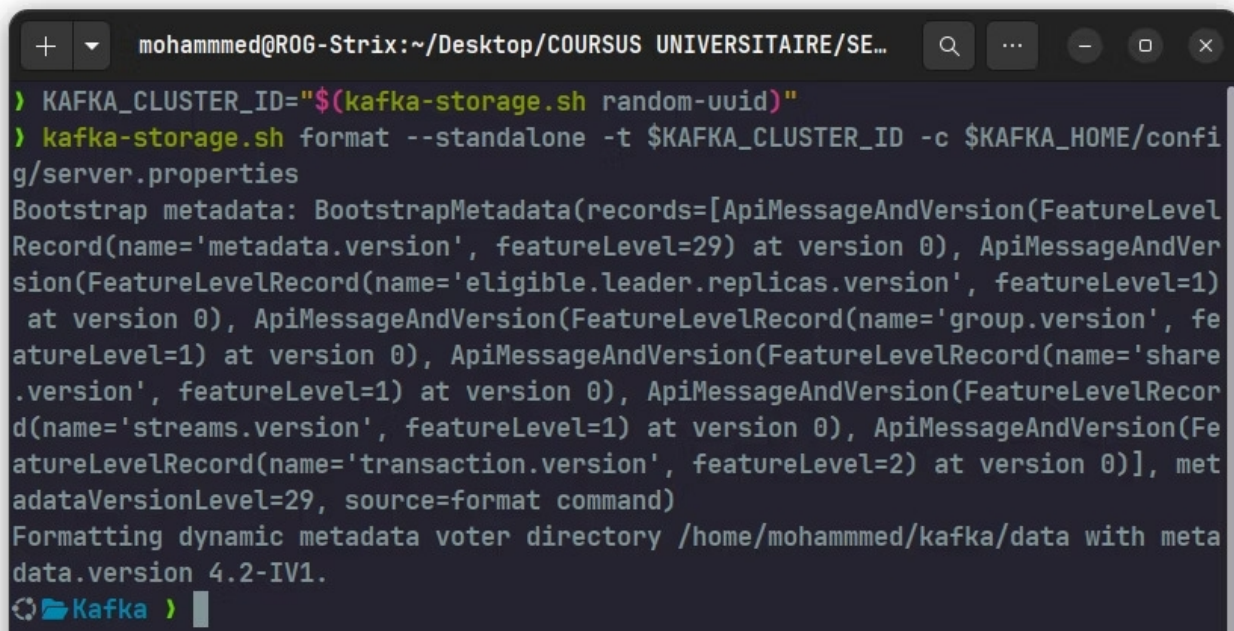
3. Configuration du fichier de serveur config/server.properties



The screenshot shows a terminal window with a text editor open to the file `config/server.properties`. The editor's title bar reads `nvim config/server.properties`. The file content includes configurations for socket buffers, log directories, and the number of partitions. The status bar at the bottom indicates the editor is in `INSERT` mode and the file is `server.properties` at line 76 of 15.

```
61 socket.send.buffer.bytes=102400
62
63 # The receive buffer (SO_RCVBUF) used by the socket server
64 socket.receive.buffer.bytes=102400
65
66 # The maximum size of a request that the socket server will accept (protection against OOM)
67 socket.request.max.bytes=104857600
68
69
70 ##### Log Basics #####
71
72 # A comma separated list of directories under which to store log files
73 log.dirs=/home/mohammed/kafka/data
74 # The default number of log partitions per topic. More partitions allow greater
75 # parallelism for consumption, but this will also result in more files across
76 # the brokers.
77 num.partitions=1
78
79 # The number of threads per data directory to be used for log recovery @@@
```

4. installation du cluster et formatage du stockage a travers le id généré



The screenshot shows a terminal window with the following commands and output:

```
> KAFKA_CLUSTER_ID="$(kafka-storage.sh random-uuid)"
> kafka-storage.sh format --standalone -t $KAFKA_CLUSTER_ID -c $KAFKA_HOME/config/server.properties
Bootstrap metadata: BootstrapMetadata(records=[ApiMessageAndVersion(FutureLevelRecord(name='metadata.version', featureLevel=29) at version 0), ApiMessageAndVersion(FutureLevelRecord(name='eligible.leader.replicas.version', featureLevel=1) at version 0), ApiMessageAndVersion(FutureLevelRecord(name='group.version', featureLevel=1) at version 0), ApiMessageAndVersion(FutureLevelRecord(name='share.version', featureLevel=1) at version 0), ApiMessageAndVersion(FutureLevelRecord(name='streams.version', featureLevel=1) at version 0), ApiMessageAndVersion(FutureLevelRecord(name='transaction.version', featureLevel=2) at version 0)], metadataVersionLevel=29, source=format command)
Formatting dynamic metadata voter directory /home/mohammed/kafka/data with metadata.version 4.2-IV1.
Kafka >
```

5. Démarche du Broker Kafka

```
kafka-server-start.sh $KAFKA_HOME/config/server.properties
```

```
> kafka-server-start.sh $KAFKA_HOME/config/server.properties
[2026-04-30 22:19:41,963] INFO Registered `kafka:type=kafka.Log4jController` MBean (kafka.utils.Log4jControllerRegistration$)
[2026-04-30 22:19:42,202] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2026-04-30 22:19:42,204] INFO [ControllerServer id=1] Starting controller (kafka.server.ControllerServer)
[2026-04-30 22:19:42,461] INFO Updated connection-accept-rate max connection creation rate to 2147483647 (kafka.network.ConnectionQuotas)
[2026-04-30 22:19:42,492] INFO [SocketServer listenerType=CONTROLLER, nodeId=1] Created data-plane acceptor and processors for endpoint : ListenerName(CONTROLLER) (kafka.network.SocketServer)
[2026-04-30 22:19:42,498] INFO authorizerStart completed for endpoint CONTROLLER. Endpoint is now READY. (org.apache.kafka.server.network.EndpointReadyFutures)
[2026-04-30 22:19:42,499] INFO [SharedServer id=1] Starting SharedServer (kafka.server.SharedServer)
[2026-04-30 22:19:42,540] INFO [LogLoader partition=__cluster_metadata-0, dir=/home/mohammed/kafka/data] Loading producer state till offset 0 (org.apache.kafka.storage.internals.log.UnifiedLog)
[2026-04-30 22:19:42,541] INFO [LogLoader partition=__cluster_metadata-0, dir=/home/mohammed/kafka/data] Reloading from producer snapshot and rebuilding producer state from offset 0 (org.apache.kafka.storage.internals.log.UnifiedLog)
[2026-04-30 22:19:42,541] INFO [LogLoader partition=__cluster_metadata-0, dir=/home/mohammed/kafka/data] Producer state recovery took 0ms for snapshot load and
```

6. Création d'un topic

```
kafka-topics.sh --create --topic test-topic --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic test-topic.
Kafka >
```


7. Verifier les topics

a. Lister les topics

```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...
kafka-server-start.sh $KAFKA_HOME/...
> kafka-topics.sh --list --bootstrap-server localhost:9092

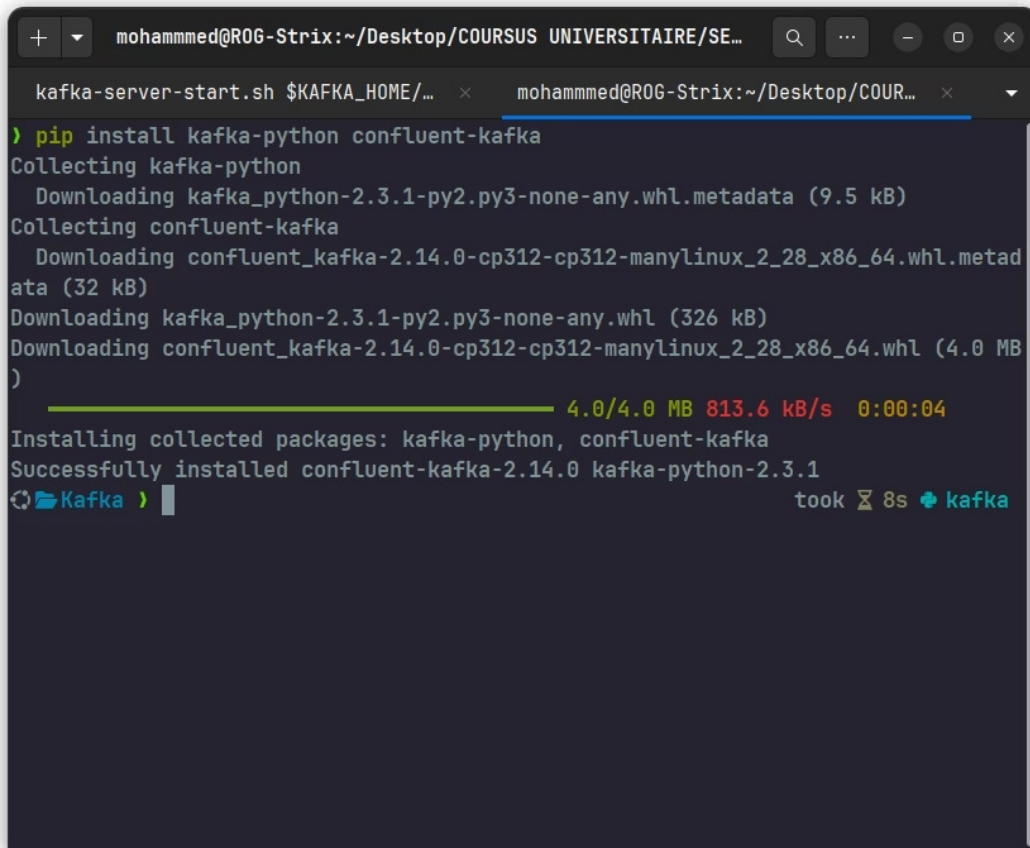
test-topic
>
Kafka > 
```

b.décrire topics

```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...
kafka-server-start.sh $KAFKA_HOME/...
> kafka-topics.sh --describe --topic test-topic --bootstrap-server localhost:9092

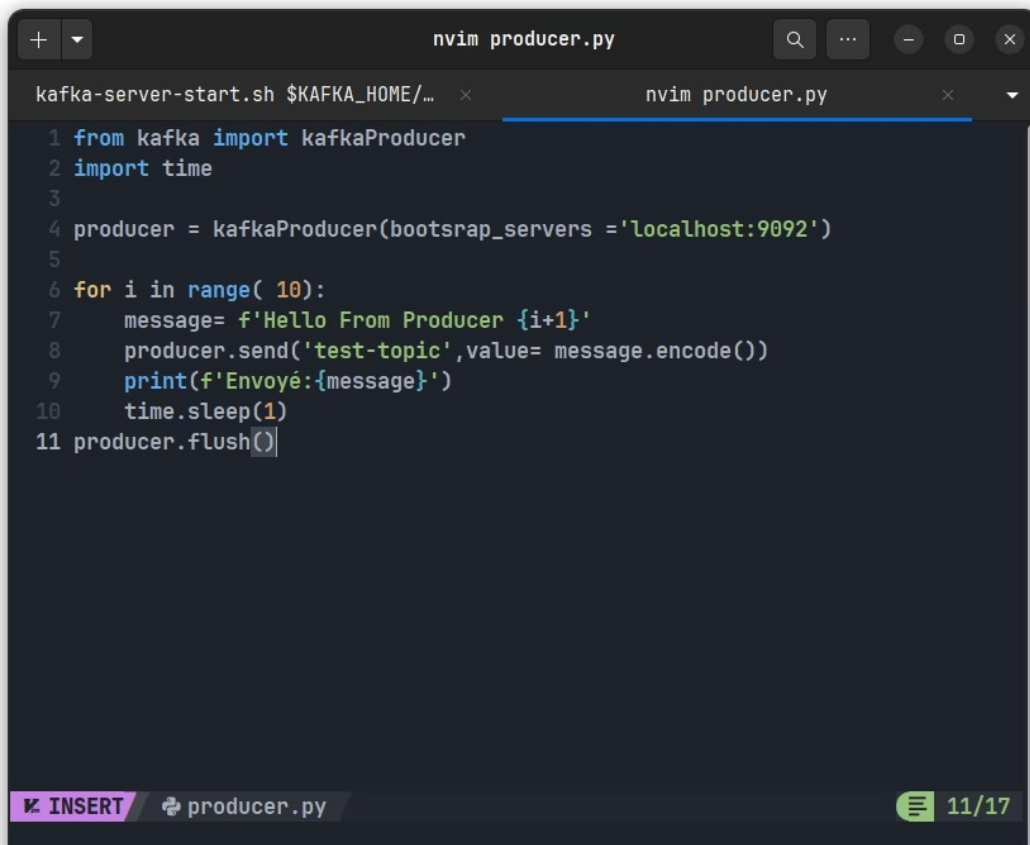
Topic: test-topic      TopicId: RCivt0l4RpaRVPaUgv9JDg PartitionCount: 1      R
eplicationFactor: 1    Configs: min.insync.replicas=1,segment.bytes=1073741824
      Topic: test-topic Partition: 0    Leader: 1      Replicas: 1    I
sr: 1    Elr:      LastKnownElr:
Kafka > 
```

8. Installation du du Kafka python



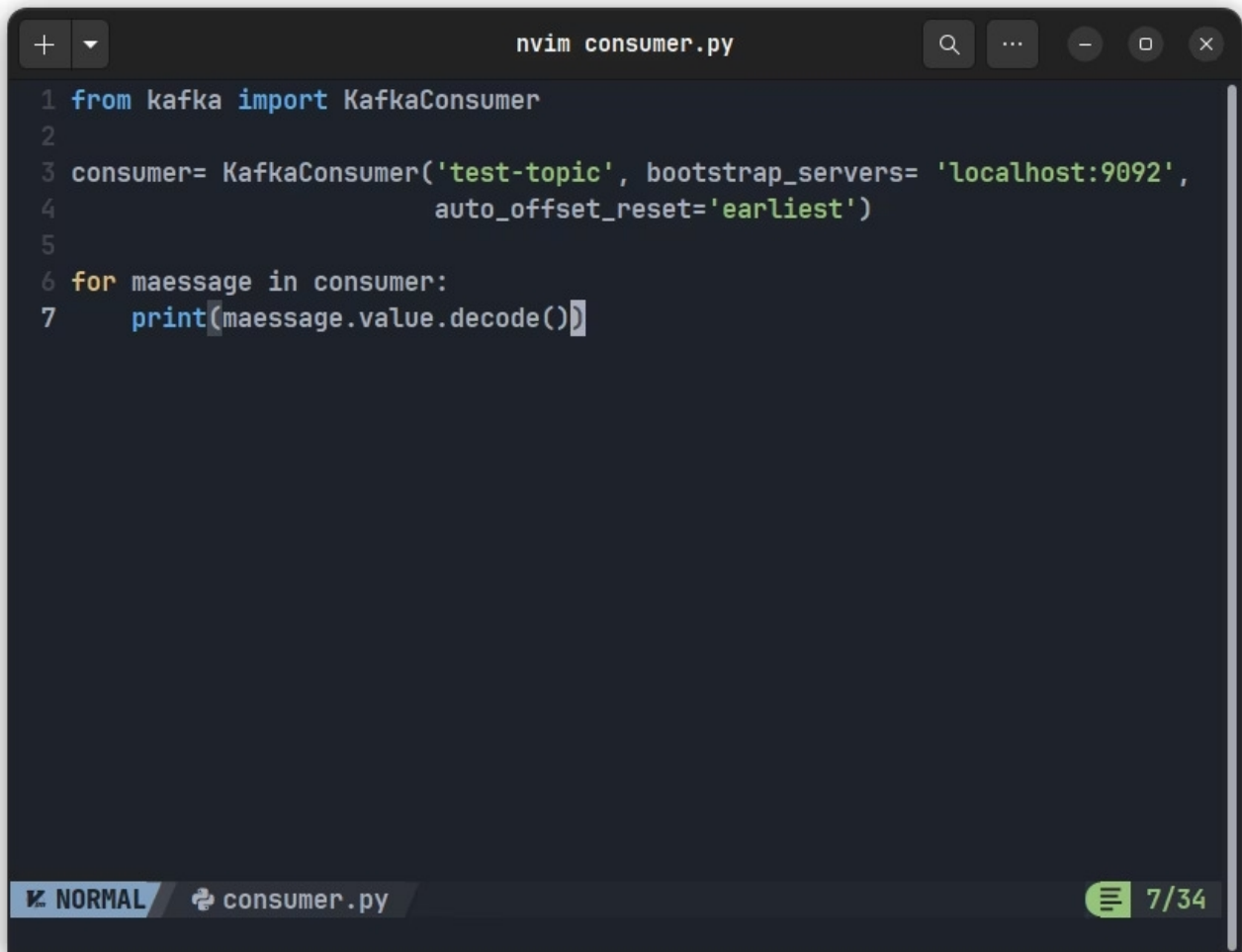
```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...
kafka-server-start.sh $KAFKA_HOME/... x mohammed@ROG-Strix:~/Desktop/COUR... x
> pip install kafka-python confluent-kafka
Collecting kafka-python
  Downloading kafka_python-2.3.1-py2.py3-none-any.whl.metadata (9.5 kB)
Collecting confluent-kafka
  Downloading confluent_kafka-2.14.0-cp312-cp312-manylinux_2_28_x86_64.whl.metad
ata (32 kB)
Downloading kafka_python-2.3.1-py2.py3-none-any.whl (326 kB)
Downloading confluent_kafka-2.14.0-cp312-cp312-manylinux_2_28_x86_64.whl (4.0 MB
)
 4.0/4.0 MB 813.6 kB/s 0:00:04
Installing collected packages: kafka-python, confluent-kafka
Successfully installed confluent-kafka-2.14.0 kafka-python-2.3.1
🔄 Kafka > took 8s kafka
```

9. Crétion du producer pyrhon



```
nvim producer.py
kafka-server-start.sh $KAFKA_HOME/... x nvim producer.py x
1 from kafka import kafkaProducer
2 import time
3
4 producer = kafkaProducer(bootstrap_servers = 'localhost:9092')
5
6 for i in range( 10):
7     message= f'Hello From Producer {i+1}'
8     producer.send('test-topic',value= message.encode())
9     print(f'Envoyé:{message}')
10    time.sleep(1)
11 producer.flush()
```

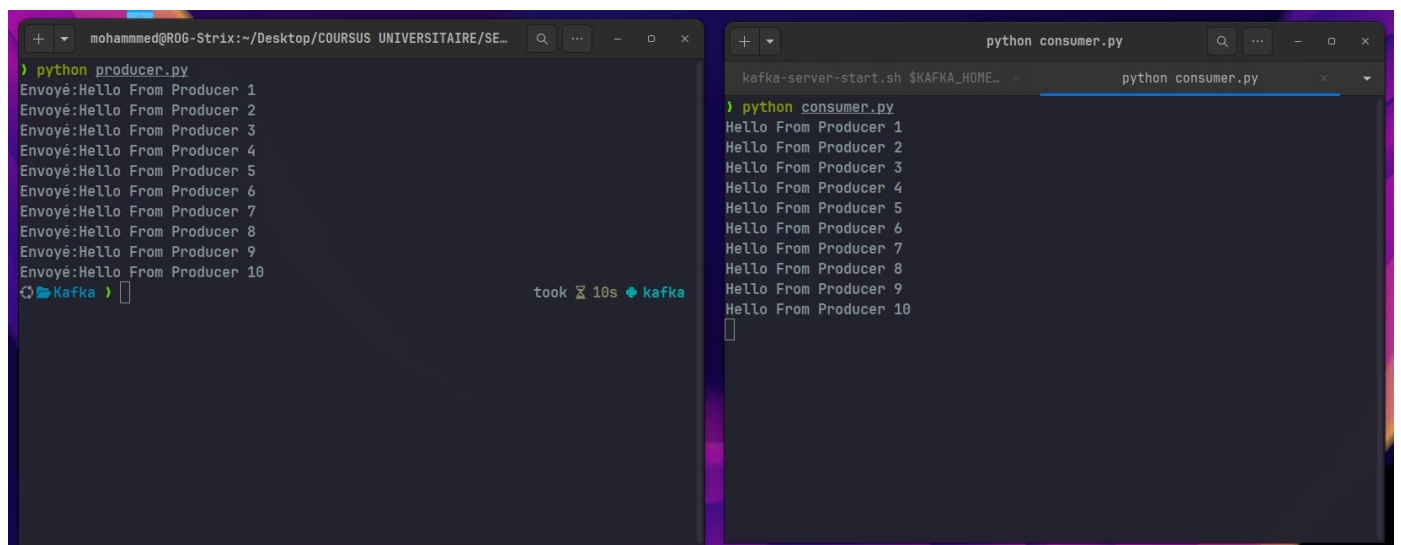
consumer python



```
nvim consumer.py
1 from kafka import KafkaConsumer
2
3 consumer= KafkaConsumer('test-topic', bootstrap_servers= 'localhost:9092',
4                           auto_offset_reset='earliest')
5
6 for message in consumer:
7     print(message.value.decode())
```

NORMAL consumer.py 7/34

Résultat finale



```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...
> python producer.py
Envoyé:Hello From Producer 1
Envoyé:Hello From Producer 2
Envoyé:Hello From Producer 3
Envoyé:Hello From Producer 4
Envoyé:Hello From Producer 5
Envoyé:Hello From Producer 6
Envoyé:Hello From Producer 7
Envoyé:Hello From Producer 8
Envoyé:Hello From Producer 9
Envoyé:Hello From Producer 10
Kafka > took 10s + kafka
```

```
python consumer.py
kafka-server-start.sh $KAFKA_HOME...
python consumer.py
> python consumer.py
Hello From Producer 1
Hello From Producer 2
Hello From Producer 3
Hello From Producer 4
Hello From Producer 5
Hello From Producer 6
Hello From Producer 7
Hello From Producer 8
Hello From Producer 9
Hello From Producer 10
```